

L J INSTITUTE OF COMPUTER APPLICATIONS
Master of Computer Applications (MCA)
Syllabus for Batch - 2025
Semester: III

Category	Discipline Specific Core -I			
Course Title	Web Application Development Using DOT NET(WAD)			
Scheme and Credits	Theory	Tutorial	Lab	Credits
	3	0	4	5
Pre-requisites (if any)	The core concepts of HTML, and OOP (Object Oriented Programming).			

1. Course Objectives:

CO1	Understand the architecture and components of the .NET platform and apply object-oriented programming principles to develop robust software applications.
CO2	Design and implement interactive, user-centric web applications by integrating user interface controls, event handling, validation mechanisms, state management, and reusable application components.
CO3	Apply modern programming techniques, including collections, LINQ, and data processing methods, to efficiently manage, query, and manipulate application data.
CO4	Develop data-driven enterprise applications by integrating databases, implementing data access strategies, and presenting information through appropriate user interfaces and architectural patterns.
CO5	Build intelligent and scalable software solutions using MVC architecture and AI-powered services to address real-world application requirements and enhance user experiences.

2. Course Contents:

Unit	Course Content	Weightage
Unit I	Introduction Framework and Overview of Controls. • Overview of ASP.NET framework, architecture, the architecture of CLR, • History of C#, Versions and Features, Procedural Programming Approach • Framework Class Library (FCL), JIT and Types of JIT, and Manage and Unmanaged code.	20%

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	- Handling page events, IsPostBack properties. - The oops concept in C#.NET. The class, properties, and implemented properties. AI Integration: - Smart Error Helper	
Unit II	Using the Standard Controls. - Properties, Methods and Events, Working with Form Control - Standard Controls, Validation Controls, Data Controls, Navigation Controls, Login Controls, HTML Controls - Label, Literal, Textbox, Button, Link Button, Image Button, and handling command events, Checkbox, Radiobutton, Checked ListBox Controls. - Image, Panel and Hyperlink control. - Dropdown list, Radiobuttonlist, listbox, Checkbox list, Bullet list control. - Creating Menu's and Menu Item's AI Integration - Introduction to OpenAI API integration in .NET	20%
Unit III	Validation Controls, MasterPages, and State Management. - RequiredField Validator, Range Validator, Compare Validator, RegularExpression Validator, Custom Validator, Validation Summary. - Master Pages - Application state (Cookies and Session). - LINQ introduction, LINQ Query Syntax, LINQ Method Syntax, Query Operator: Where, Select, OrderBy, OrderByDescending, Aggregate, Average, Count, Max, Min, Sum, GroupBy, First, FirstOrDefault, Last, LastOrDefault, Single, SingleOrDefault. - The concept of tuple. The List collection.	20%

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	AI Integration: • Intelligent validation • Chatbot implementation • AI Search Assistant with LINQ	
Unit IV	ADO.NET Fundamental • Introduction To ADO.NET Technology • The Gridview, DataList, and Form view controls with sorting and paging events. • Database access without ADO.NET, The ADO.NET Architecture, ADO.NET Data Provider. Fundamental ADO.NET Classes, Connection class, Connection String, instance Connection, Calling Stored Procedure. AI Integration • AI FAQ Assistant • Natural Language SQL Query App	20%
Unit V	MVC.Net (MVC.Net Framework). • The Overview of MVC architecture, • Controller: Controller Basics, Home Controller, Creating New Controllers, Write Action Methods, Parameters in Controller Actions, Routing: URL, Introduction to Routing, Defining Route, Named Routes. • The View: The Purpose of views, Viewdata, Viewbag and TempData, Adding View, The Razor View Engines: Introduction Razor View Engines, Razor Code Expressions, Razor Syntax: Implicit Code Expression, Explicit Code Expression, Code Block, Combining Text and Markup. • The Model: The basics of the Model, How to return the model from the controller by using the Action Method, and How to use the model in the View. AI Integration: • AI Travel Planner • AI Resume Analyzer	20%

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Program Outcomes (POs)

PO Code	Program Outcomes
PO1	Computational Knowledge
PO2	Problem Analysis
PO3	Design and Development of Solutions
PO4	Modern Tool Usage
PO5	Individual and Team Work
PO6	Professional Ethics
PO7	Communication Skills
PO8	Life-long Learning

CO-PO Mapping Matrix

Mapping Scale:

- 3 = High
- 2 = Moderate
- 1 = Low

COs \ POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	2	2	1	-	-	-	1
CO2.	2	2	3	2	1	-	1	1
CO3	3	3	2	2	-	-	-	1
CO4	2	3	3	3	1	-	1	1
CO5	2	3	3	3	1	1	1	2

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Reference Books:

1. ASP.NET 4 Unleashed, Stephen Walther, Kevin Hoffman, Nate Dudek.
2. Professional ASP.NET MVC 4 By Jon Galloway, Phil Hack, Brad Wilson, K. Scott Allen, WROX Publication
3. Pro ASP.NET 4 in C# 2010, Matthew MacDonald, Adam Freeman and Mario Szpuszta, 4th edition, Apress
4. Beginning ASP.NET 2.0 Database Beta Preview, John Kauffman with Thiry Thangarathinam, Wrox
5. ASP.NET Web Developers Guide, Mesbah Ahmed, Chris Garrent, Jerney Faircloth, Chris Payne, Global Knowledge.